# STAT 423 Winter 2026 - Final Project

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```r
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating=orig_data$overall_rating, age=orig_data$fr

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified", orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                             ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                                    ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                                           ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                                                  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "O

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy, danceability = orig_dat


#Wind
cleaned_data$has_winds <- ifelse(orig_data$clarinet, "True",
                                 ifelse(orig_data$flute_piccolo, "True",
                                        ifelse(orig_data$harmonica, "True",
                                               ifelse(orig_data$kazoo, "True",
                                                      ifelse(orig_data$ocarina, "True",
                                                             ifelse(orig_data$saxophone, "True",
                                                            ifelse(orig_data$accordion, "True", "False")



#Brass
cleaned_data$has_brass <- ifelse(orig_data$trumpet, "True", "False")

#Percussion
cleaned_data$has_percussion <- ifelse(orig_data$piano_keyboard_based, "True",
                                 ifelse(orig_data$bongos, "True",
                                        ifelse(orig_data$cowbell, "True",
                                               ifelse(orig_data$handclaps_snaps, "True", "False"))))

#Vocal or Whistle
cleaned_data$has_vocal_or_whistle <- ifelse(orig_data$vocally_based, "True",
                                 ifelse(orig_data$falsetto_vocal, "True",
                                        ifelse(orig_data$human_whistling, "True", "False")))
```

```r
#String
cleaned_data$has_strings <- ifelse(orig_data$bass_based, "True",
                                ifelse(orig_data$guitar_based, "True",
                                       ifelse(orig_data$orchestral_strings, "True",
                                              ifelse(orig_data$banjo, "True",
                                                     ifelse(orig_data$mandolin, "True",
                                                            ifelse(orig_data$pedal_lap_steel, "True",
                                                             ifelse(orig_data$sitar, "True",
                                                             ifelse(orig_data$ukulele, "True",
                                                             ifelse(orig_data$violin, "True","False"))


cleaned_data$year <- as.numeric(substr(orig_data$date, 1, unlist(gregexpr("-", orig_data$date))[1] - 1)

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

cutoff <- round(min(cleaned_data$year) + ((max(cleaned_data$year) - min(cleaned_data$year)) / 2))

first_subpop <- cleaned_data[cleaned_data$year < cutoff, ]

second_subpop <- cleaned_data[cleaned_data$year >= cutoff, ]
```

```r
model_subpop1_1 <- lm(data = first_subpop, rating ~ age + race + gender + genre + energy + danceability
summary(model_subpop1_1)
```

```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = first_subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.7774 -1.2578 -0.0063  1.3111  4.4504
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.012405   0.825236   9.709  < 2e-16 ***
## age             -0.030320   0.010595  -2.862  0.00433 **
## raceWhite       -0.648978   0.163236  -3.976 7.72e-05 ***
## genderNot Male   0.001685   0.155477   0.011  0.99135
## genreJazz        1.046241   0.852095   1.228  0.21990
## genreOther       0.463522   0.413876   1.120  0.26310
## genrePop        -0.180256   0.386269  -0.467  0.64088
## genreRock        0.526001   0.394097   1.335  0.18239
## energy          -0.008542   0.005863  -1.457  0.14555
## danceability    -0.013090   0.005734  -2.283  0.02271 *
## happiness        0.006972   0.003926   1.776  0.07620 .
## bpm              0.001554   0.002986   0.520  0.60295
## loudness_db      0.063151   0.030448   2.074  0.03842 *
## length_sec       0.002120   0.001475   1.437  0.15111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.856 on 729 degrees of freedom
## Multiple R-squared:  0.07473,    Adjusted R-squared:  0.05823
## F-statistic: 4.529 on 13 and 729 DF,  p-value: 1.76e-07
```

```r
model_subpop1_2 <- lm(data = first_subpop, rating ~ age + gender + race + age*genre + age*energy + age*d
summary(model_subpop1_2)
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = first_subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.8819 -1.2115 -0.0254  1.2281  4.4727
##
## Coefficients: (1 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 1.122e+01  4.075e+00   2.753  0.00606 **
## age                        -1.025e-01  1.180e-01  -0.868  0.38556
## genderNot Male              9.917e-02  1.661e+00   0.060  0.95241
## raceWhite                  -2.046e+00  2.079e+00  -0.984  0.32545
## genreJazz                  -2.095e-02  3.139e+00  -0.007  0.99468
## genreOther                 -1.395e+00  2.554e+00  -0.546  0.58513
## genrePop                   -1.555e+00  2.455e+00  -0.634  0.52657
## genreRock                  -9.519e-01  2.531e+00  -0.376  0.70695
## energy                      7.785e-03  2.797e-02   0.278  0.78085
## danceability               -6.778e-02  3.036e-02  -2.233  0.02588 *
## happiness                   2.095e-02  1.934e-02   1.083  0.27903
## bpm                         5.520e-03  1.455e-02   0.379  0.70452
## loudness_db                 1.118e-01  1.460e-01   0.766  0.44420
## length_sec                 -9.933e-04  7.830e-03  -0.127  0.89909
## age:genreJazz              -1.055e-02  7.959e-02  -0.133  0.89456
## age:genreOther              5.163e-02  6.936e-02   0.744  0.45691
## age:genrePop                8.418e-03  6.482e-02   0.130  0.89672
## age:genreRock              -2.577e-03  6.760e-02  -0.038  0.96960
## age:energy                 -9.549e-04  9.949e-04  -0.960  0.33748
## age:danceability            1.781e-03  9.875e-04   1.803  0.07179 .
## age:happiness              -2.467e-04  6.395e-04  -0.386  0.69983
## age:bpm                     9.001e-05  4.617e-04   0.195  0.84550
## age:loudness_db            -6.034e-04  4.927e-03  -0.122  0.90256
## age:length_sec              5.662e-05  2.685e-04   0.211  0.83308
## raceWhite:genreJazz         1.873e+00  2.108e+00   0.889  0.37447
## raceWhite:genreOther        4.061e-01  1.469e+00   0.277  0.78220
## raceWhite:genrePop          1.540e+00  1.439e+00   1.070  0.28488
## raceWhite:genreRock         1.960e+00  1.479e+00   1.325  0.18546
## raceWhite:energy            1.533e-02  1.287e-02   1.191  0.23407
```

```
## raceWhite:danceability      -5.342e-03  1.251e-02  -0.427  0.66952
## raceWhite:happiness         -4.542e-03  8.705e-03  -0.522  0.60200
## raceWhite:bpm               -1.108e-02  6.886e-03  -1.609  0.10817
## raceWhite:loudness_db       -2.007e-02  6.400e-02  -0.314  0.75390
## raceWhite:length_sec         4.570e-03  3.333e-03   1.371  0.17078
## genderNot Male:genreJazz            NA         NA      NA       NA
## genderNot Male:genreOther   -1.664e+00  9.745e-01  -1.708  0.08810 .
## genderNot Male:genrePop     -1.098e+00  9.203e-01  -1.193  0.23330
## genderNot Male:genreRock    -1.412e+00  9.677e-01  -1.460  0.14487
## genderNot Male:energy        2.062e-03  1.381e-02   0.149  0.88132
## genderNot Male:danceability  3.385e-02  1.269e-02   2.667  0.00783 **
## genderNot Male:happiness    -1.285e-02  8.571e-03  -1.499  0.13424
## genderNot Male:bpm           3.439e-03  6.717e-03   0.512  0.60886
## genderNot Male:loudness_db  -4.889e-02  6.973e-02  -0.701  0.48346
## genderNot Male:length_sec   -4.775e-03  3.415e-03  -1.398  0.16246
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.838 on 700 degrees of freedom
## Multiple R-squared:  0.1286, Adjusted R-squared:  0.07634
## F-statistic:  2.46 on 42 and 700 DF,  p-value: 1.632e-06
```

```r
model_subpop2_1 <- lm(data = second_subpop, rating ~ age + race + gender + genre + energy + danceability
summary(model_subpop2_1)
```

```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = second_subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3964 -1.0558  0.0392  0.9655  4.5133
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     5.016752   1.960993   2.558  0.01089 *
## age            -0.001924   0.012044  -0.160  0.87313
## raceWhite      -0.128474   0.187366  -0.686  0.49331
## genderNot Male  0.484635   0.162164   2.989  0.00298 **
## genreOther      0.707159   1.615313   0.438  0.66178
## genrePop        0.748135   1.619543   0.462  0.64438
## genreRock       0.657571   1.637667   0.402  0.68825
## energy         -0.015693   0.007433  -2.111  0.03537 *
## danceability    0.003811   0.007058   0.540  0.58957
## happiness       0.011840   0.004400   2.691  0.00742 **
## bpm            -0.001863   0.003034  -0.614  0.53954
## loudness_db     0.081621   0.050192   1.626  0.10470
## length_sec      0.004682   0.001934   2.420  0.01595 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.595 on 398 degrees of freedom
```

```
## Multiple R-squared:  0.07127,	Adjusted R-squared:  0.04327
## F-statistic: 2.545 on 12 and 398 DF,  p-value: 0.002998
```

```r
model_subpop2_2 <- lm(data = second_subpop, rating ~ age + gender + race +age*genre + age*energy + age*
summary(model_subpop2_2)
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = second_subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.2473 -1.0138  0.0664  0.9589  3.6915
##
## Coefficients: (3 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                -1.962e+00  4.880e+00  -0.402  0.68790
## age                         2.331e-01  1.437e-01   1.622  0.10573
## genderNot Male             -9.228e-01  2.197e+00  -0.420  0.67474
## raceWhite                  -1.117e+00  2.418e+00  -0.462  0.64441
## genreOther                  4.728e+00  2.903e+00   1.628  0.10428
## genrePop                    6.778e+00  2.969e+00   2.283  0.02300 *
## genreRock                   1.609e+00  1.681e+00   0.957  0.33913
## energy                      1.072e-02  2.993e-02   0.358  0.72044
## danceability                1.117e-02  3.239e-02   0.345  0.73033
## happiness                  -6.277e-03  1.940e-02  -0.324  0.74642
## bpm                        -1.537e-02  1.478e-02  -1.040  0.29899
## loudness_db                -8.527e-03  2.325e-01  -0.037  0.97076
## length_sec                  1.609e-02  9.603e-03   1.676  0.09460 .
## age:genreOther             -9.014e-02  5.513e-02  -1.635  0.10289
## age:genrePop               -1.817e-01  5.882e-02  -3.089  0.00216 **
## age:genreRock                      NA         NA      NA       NA
## age:energy                 -1.092e-03  9.919e-04  -1.100  0.27183
## age:danceability            4.116e-05  1.026e-03   0.040  0.96802
## age:happiness               6.305e-04  6.349e-04   0.993  0.32131
## age:bpm                     3.916e-04  4.937e-04   0.793  0.42812
## age:loudness_db             4.001e-03  7.673e-03   0.521  0.60238
## age:length_sec             -5.426e-04  3.175e-04  -1.709  0.08824 .
## raceWhite:genreOther       -6.881e-01  1.096e+00  -0.628  0.53061
## raceWhite:genrePop          6.208e-02  1.107e+00   0.056  0.95533
## raceWhite:genreRock                NA         NA      NA       NA
## raceWhite:energy           -2.158e-02  1.578e-02  -1.368  0.17215
## raceWhite:danceability      1.540e-02  1.591e-02   0.968  0.33351
## raceWhite:happiness        -6.471e-03  1.026e-02  -0.631  0.52846
## raceWhite:bpm               1.194e-02  6.783e-03   1.760  0.07915 .
## raceWhite:loudness_db       1.438e-01  1.048e-01   1.372  0.17094
## raceWhite:length_sec        7.067e-03  4.177e-03   1.692  0.09149 .
```

```
## genderNot Male:genreOther   -1.591e-01  8.633e-01  -0.184  0.85384
## genderNot Male:genrePop      1.766e-01  9.150e-01   0.193  0.84703
## genderNot Male:genreRock            NA         NA      NA       NA
## genderNot Male:energy        2.686e-02  1.522e-02   1.765  0.07845 .
## genderNot Male:danceability -2.706e-02  1.442e-02  -1.877  0.06128 .
## genderNot Male:happiness     2.947e-03  9.101e-03   0.324  0.74625
## genderNot Male:bpm          -1.735e-04  6.152e-03  -0.028  0.97752
## genderNot Male:loudness_db  -1.377e-01  1.032e-01  -1.334  0.18311
## genderNot Male:length_sec    2.612e-03  4.072e-03   0.641  0.52162
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.568 on 374 degrees of freedom
## Multiple R-squared:  0.1563, Adjusted R-squared:  0.0751
## F-statistic: 1.925 on 36 and 374 DF,  p-value: 0.001511
```

```r
model_total_1 <- lm(data = cleaned_data, rating ~ age + race + gender + genre + energy + danceability +
summary(model_total_1)
```

```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = cleaned_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.8024 -1.2244 -0.0203  1.2248  4.5071
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.0722615  0.6747344  10.482  < 2e-16 ***
## age             -0.0235027  0.0080080  -2.935 0.003403 **
## raceWhite       -0.4425431  0.1231832  -3.593 0.000341 ***
## genderNot Male   0.2008111  0.1139569   1.762 0.078309 .
## genreJazz        1.1915765  0.8098788   1.471 0.141485
## genreOther       0.2896938  0.3717489   0.779 0.435981
## genrePop        -0.0309389  0.3602833  -0.086 0.931582
## genreRock        0.5806698  0.3666694   1.584 0.113555
## energy          -0.0086781  0.0045002  -1.928 0.054054 .
## danceability    -0.0098823  0.0043489  -2.272 0.023249 *
## happiness        0.0103522  0.0028992   3.571 0.000371 ***
## bpm             -0.0006552  0.0021639  -0.303 0.762097
## loudness_db      0.0442563  0.0223927   1.976 0.048353 *
## length_sec       0.0027224  0.0011510   2.365 0.018188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.78 on 1140 degrees of freedom
## Multiple R-squared:  0.05086,    Adjusted R-squared:  0.04003
## F-statistic: 4.699 on 13 and 1140 DF,  p-value: 5.732e-08
```

```r
model_total_2 <- lm(data = cleaned_data, rating ~ age + gender + race +age*genre + age*energy + age*danc
summary(model_total_2)
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = cleaned_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.7534 -1.1493 -0.0184  1.1834  4.4488
##
## Coefficients: (1 not defined because of singularities)
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                9.492e+00  3.315e+00   2.863  0.00427 **
## age                       -6.391e-02  9.526e-02  -0.671  0.50243
## genderNot Male             3.706e-01  1.367e+00   0.271  0.78633
## raceWhite                 -1.378e+00  1.751e+00  -0.787  0.43120
## genreJazz                  4.795e-01  2.932e+00   0.164  0.87014
## genreOther                -1.873e+00  2.329e+00  -0.804  0.42133
## genrePop                  -1.092e+00  2.295e+00  -0.476  0.63416
## genreRock                 -8.983e-01  2.346e+00  -0.383  0.70184
## energy                     1.182e-02  2.016e-02   0.586  0.55775
## danceability              -3.883e-02  2.153e-02  -1.803  0.07165 .
## happiness                  1.270e-02  1.342e-02   0.946  0.34425
## bpm                       -3.741e-03  1.020e-02  -0.367  0.71388
## loudness_db                4.239e-02  1.071e-01   0.396  0.69223
## length_sec                 1.714e-03  5.632e-03   0.304  0.76091
## age:genreJazz             -2.004e-02  7.403e-02  -0.271  0.78671
## age:genreOther             4.622e-02  6.149e-02   0.752  0.45237
## age:genrePop              -1.034e-02  6.027e-02  -0.172  0.86385
## age:genreRock             -1.651e-03  6.219e-02  -0.027  0.97883
## age:energy                -8.336e-04  6.954e-04  -1.199  0.23091
## age:danceability           9.308e-04  6.870e-04   1.355  0.17574
## age:happiness              1.533e-04  4.417e-04   0.347  0.72865
## age:bpm                    7.048e-05  3.299e-04   0.214  0.83087
## age:loudness_db           -1.306e-03  3.605e-03  -0.362  0.71717
## age:length_sec            -4.083e-05  1.896e-04  -0.215  0.82959
## raceWhite:genreJazz        1.658e+00  1.996e+00   0.831  0.40631
## raceWhite:genreOther       9.835e-01  1.363e+00   0.721  0.47084
## raceWhite:genrePop         1.669e+00  1.355e+00   1.232  0.21811
## raceWhite:genreRock        1.991e+00  1.384e+00   1.439  0.15043
## raceWhite:energy           7.483e-04  9.274e-03   0.081  0.93570
## raceWhite:danceability    -3.031e-03  8.984e-03  -0.337  0.73587
## raceWhite:happiness       -7.637e-03  5.873e-03  -1.300  0.19377
## raceWhite:bpm             -3.436e-04  4.386e-03  -0.078  0.93758
## raceWhite:loudness_db      8.519e-02  4.553e-02   1.871  0.06158 .
```

```
## raceWhite:length_sec         4.248e-03  2.388e-03   1.778  0.07561 .
## genderNot Male:genreJazz            NA         NA      NA       NA
## genderNot Male:genreOther   -1.514e+00  8.832e-01  -1.714  0.08683 .
## genderNot Male:genrePop     -1.077e+00  8.687e-01  -1.240  0.21536
## genderNot Male:genreRock    -1.739e+00  9.057e-01  -1.920  0.05517 .
## genderNot Male:energy        4.152e-03  9.688e-03   0.429  0.66831
## genderNot Male:danceability  1.616e-02  9.138e-03   1.768  0.07734 .
## genderNot Male:happiness    -7.980e-03  6.116e-03  -1.305  0.19225
## genderNot Male:bpm           3.348e-03  4.539e-03   0.738  0.46084
## genderNot Male:loudness_db  -5.966e-03  4.702e-02  -0.127  0.89907
## genderNot Male:length_sec   -3.146e-04  2.412e-03  -0.130  0.89627
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.764 on 1111 degrees of freedom
## Multiple R-squared:  0.09126,    Adjusted R-squared:  0.05691
## F-statistic: 2.657 on 42 and 1111 DF,  p-value: 9.427e-08
```